

NutriStart

Building a Production-Ready Food Subscription Platform

A full-stack engineering project built to explore modern software architecture, scalable backend design, cloud deployment, authentication, CI/CD automation, and production engineering practices.

Flutter • FastAPI • Supabase • JWT Authentication

GitHub Actions • Railway • PostgreSQL

Engineering Portfolio & Technical Case Study

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This document captures the complete journey of NutriStart, from initial architecture decisions to public cloud deployment, highlighting the technical decisions, debugging experiences, and engineering lessons learned along the way.

Executive Summary

NutriStart evolved from a learning project into a publicly deployed cloud-hosted application. The goal was to gain practical experience across frontend development, backend engineering, authentication, databases, DevOps, deployment, and scalable architecture.

Project Overview

The platform supports meal browsing, subscription management, order placement, authentication, and future expansion into delivery partner workflows and analytics. The architecture was designed with scalability in mind from the early stages.

Technology Stack

Frontend: Flutter (Android, iOS & Web)

Backend: FastAPI

Database: Supabase PostgreSQL

Authentication: JWT

Notifications: Firebase Cloud Messaging

Deployment: Railway + GitHub Pages

CI/CD: GitHub Actions

Architecture Overview

The platform follows a layered architecture: Application Layer → API Layer → Database Layer → External Services. Backend services include Authentication, Users, Meals, Subscriptions, Orders, Delivery, Notifications, Payments, and Administration.

Development Roadmap

Phase 1: Foundation Setup

Phase 2: Authentication

Phase 3: Meal Management

Phase 4: Subscriptions

Phase 5: Orders & Payments

Phase 6: Delivery System

Phase 7: Notifications

Phase 8: Admin Dashboard

Phase 9: Scaling & CI/CD

Phase 10: Advanced Features

Engineering Challenges & Debugging

- Android emulator networking and localhost resolution.
- JWT token persistence and secure storage.
- GitHub secret scanning and repository cleanup.
- Git branch recovery after destructive operations.
- GitHub Actions workflow debugging.
- Flutter Web deployment under GitHub Pages base path.

Deployment Journey

Frontend deployment was automated through GitHub Actions and GitHub Pages. The FastAPI backend was deployed on Railway and connected to Supabase PostgreSQL. This enabled a complete end-to-end cloud-hosted workflow.

Key Learnings

- Architecture decisions compound over time.
- Deployment is a core software feature.
- CI/CD reduces operational friction.
- Debugging methodology matters more than speed.
- Real-world engineering extends far beyond writing code.

Future Roadmap

- Android Release & Play Store Launch
- Custom Domain
- Monitoring & Logging
- Delivery Partner Application
- Analytics Dashboard
- AI-driven Recommendations
- Security Hardening

Conclusion

NutriStart demonstrates the complete journey from local development to public deployment. The project provided hands-on exposure to frontend engineering, backend architecture, database design, cloud deployment, CI/CD automation, and production readiness practices.